

 France — Monthly Intelligence Report

v4.2

Edition 005 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

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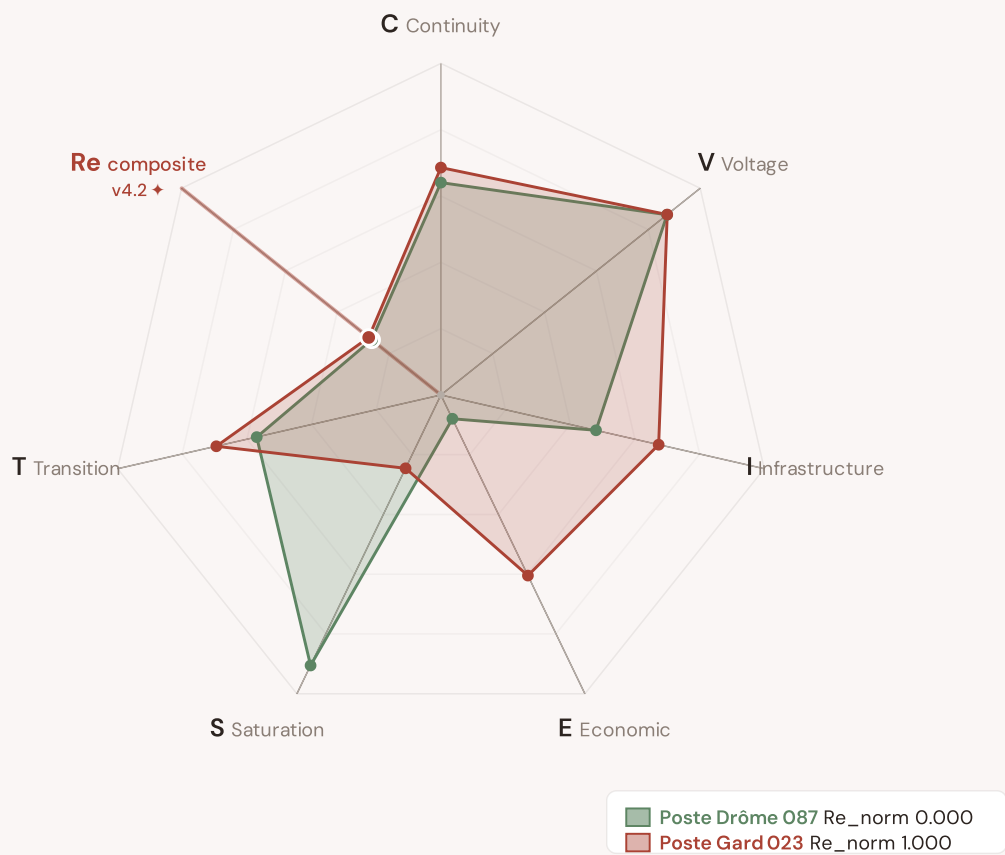
Looking Ahead
- v4.2

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). v4.2 adds **Re** — the bounded resilience composite capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)). Two worked examples overlaid: **Poste Drôme 087** (low-exposure reference, Re_norm 0.000) · **Poste Gard 023** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

7,378

938 EHV · 1,123 HV · 5,317 distribution-tier

GRID LINES MAPPED

68,379

~147,648 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 7,378 substations and 68,379 grid lines spanning ~147,648 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering France’s entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Poste Jura 033

Jura, Bourgogne-Franche-Comté · 63 kV

0.476

Medium

0.145

21th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.1561 · Re_norm 0.461

MODERATE EXPOSURE

COMPONENTS

C

Continuity: 0.549 / 0.30 (183%)

I

Infrastructure: 0.188 / 0.25 (75%)

S

Saturation: 0.289 / 0.20 (144%)

V

Voltage: 0.720 / 0.10 (720%) ▲

E

Economic: 0.059 / 0.10 (59%)

T

Transition: 0.717 / 0.05 (1433%)

MODIFIERS

R3

Consequence: 1.010 → neutral

R4

Graph Criticality: 0.977 ↓ dampens

R6a

Restoration: 1.011 ↑ amplifies

R6b

Seismic: 1.009 → neutral

R7

Digital Readiness: 1.004 → neutral

This month’s spotlight — automatically selected for July 2026 — is **Poste Jura 033** in Jura, Bourgogne-Franche-Comté. With an R_final of 0.476 (Medium band, 21th fleet percentile), its risk profile is dominated by the T (Transition) component at 1433% of its weight ceiling, followed by V (Voltage) at 720%. Modifiers collectively shift R_base by 24.3%%, reflecting the substation’s specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

3200

High/Critical + R7 < median

AVG R7 MODIFIER

1.0148

Range [0.99, 1.05]

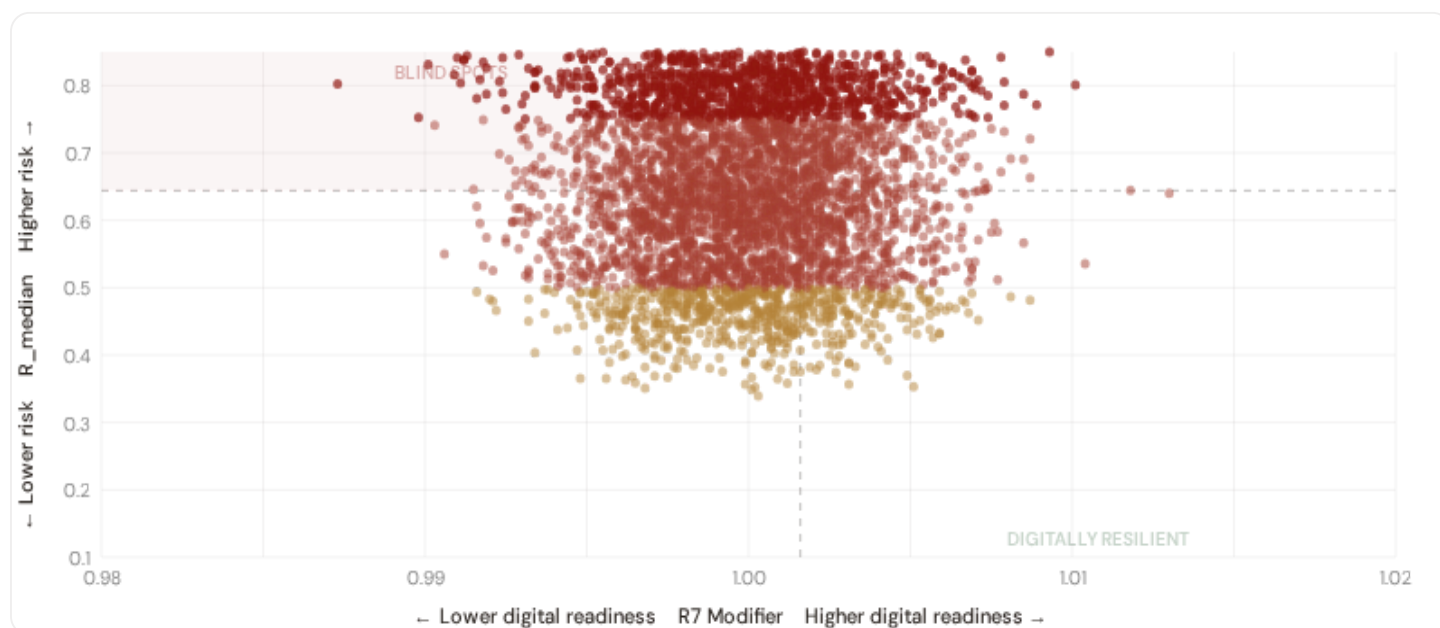
HIGH-CONSEQUENCE SUBS

2,055

27.9% · R3 ≥ 1.13

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



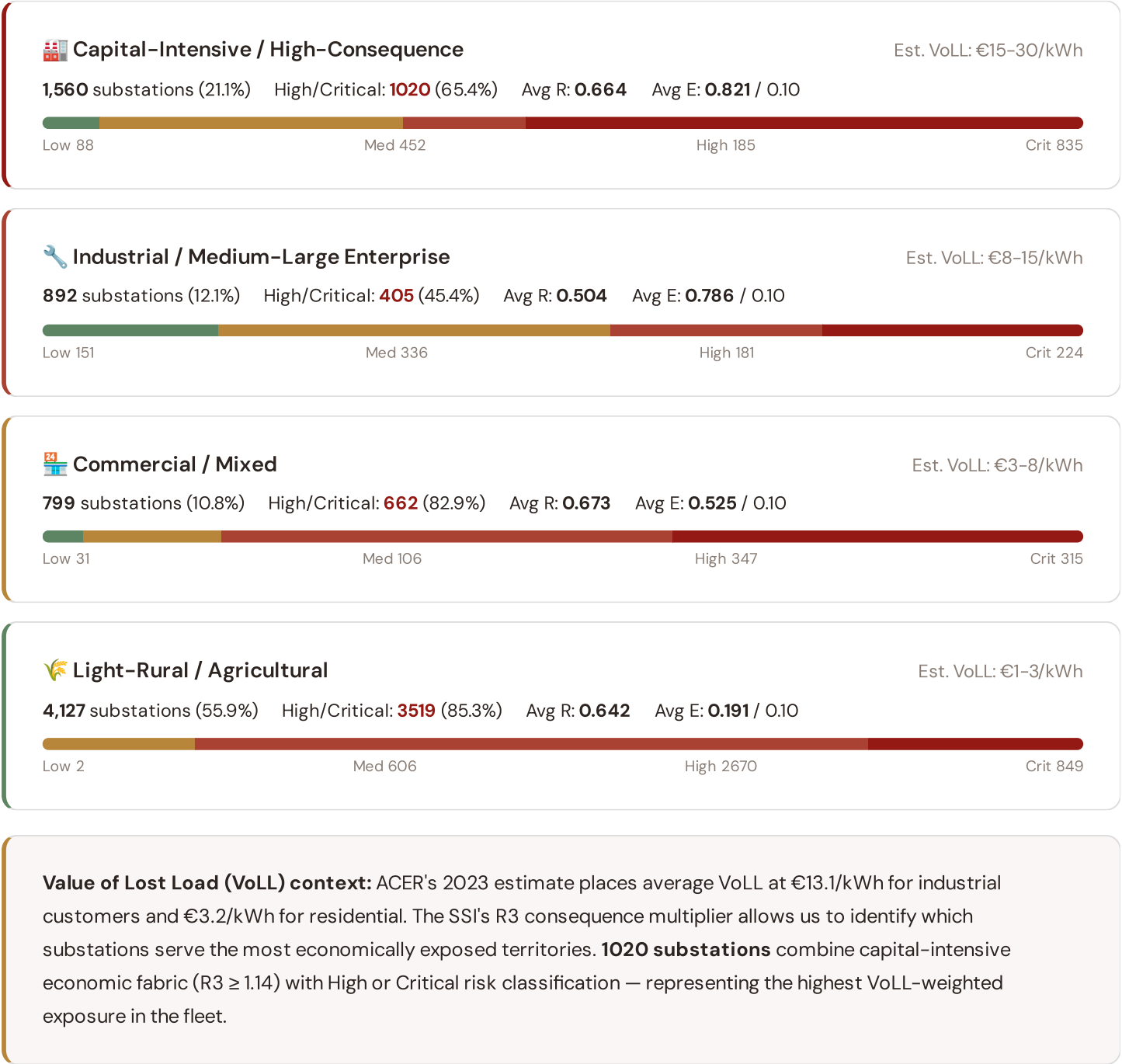
● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **3200 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0016$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Occitanie (651)**, **Nouvelle-Aquitaine (588)**, **Provence-Alpes-Côte d'Azur (388)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



The capital-intensive tier concentrates in **Île-de-France (542)**, **Auvergne-Rhône-Alpes (278)**, **Hauts-de-France (206)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 7.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Haute-Corse Δ high R	0.9894
2	Corse-du-Sud Δ high R	0.9912
3	Loir-et-Cher	0.9991
4	Moselle	0.9992
5	Indre	0.9993
6	Oise	0.9994
7	Alpes-de-Haute-Provence Δ high R	0.9994
8	Alpes-Maritimes Δ high R	0.9995
9	Finistère Δ high R	0.9995
10	Pyrénées-Atlantiques Δ high R	0.9995
11	Nord	0.9995
12	Vaucluse Δ high R	0.9996
13	Dordogne	0.9996
14	Haute-Garonne Δ high R	0.9996
15	Bouches-du-Rhône Δ high R	0.9996

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Haute-Corse** has the lowest average R7 (0.9894), while **Isère** leads at 1.0507 — a spread of 0.0613 across the R7 range. **41 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (July 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0148, median = 1.0016; 0 substations classified HIGH cyber-exposure; 3200 blind-spot substations (High/Critical risk + below-median R7); 1240 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

— variables

HOURLY — MONTHLY

High-frequency feeds

QUARTERLY / ANNUAL

Regular refresh cycle

STATIC / DERIVED



Standards + scenarios

Data Source Registry — July 2026

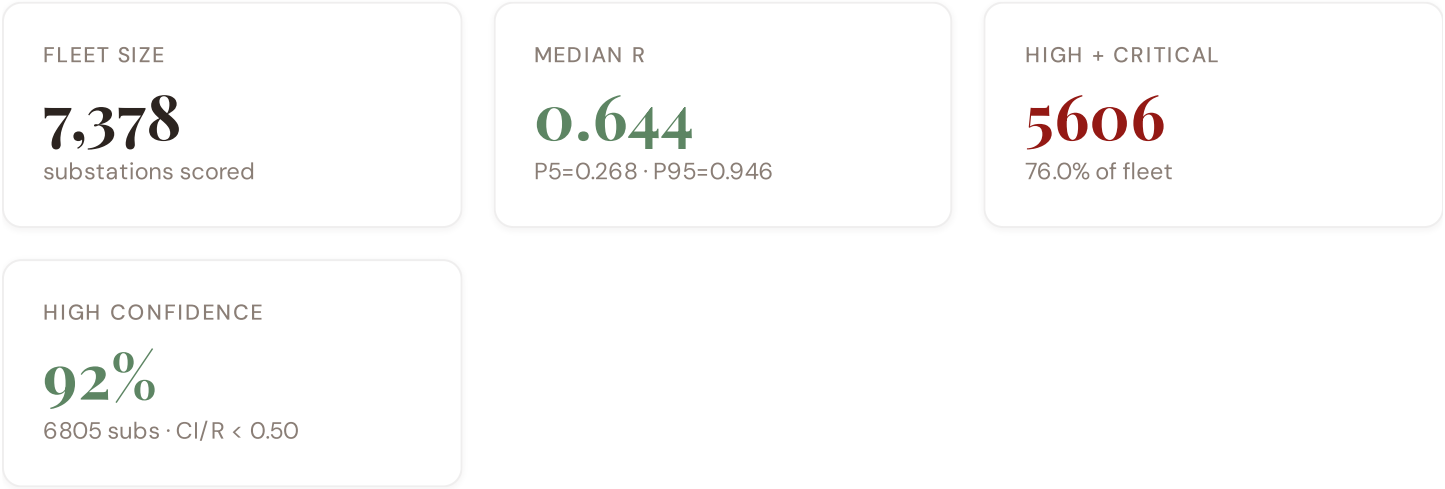
OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
RTE-N	RTE Schéma Décennal	MONTHLY	TSO zone	2 vars
DREAL	DREAL (Directions Régionales)	CONTINUOUS	Département	1 var
ODRE	ODRE Registre National (OpenDataSoft)	QUARTERLY	Commune	5 vars
SDES	SDES (Service des Données Statistiques)	QUARTERLY	Département	1 var
BPI	Bpifrance Innovation	QUARTERLY	Département	1 var
CRE	CRE (Commission de Régulation de l'Énergie)	ANNUAL	Département	8 vars
INSEE	INSEE (Institut National de la Statistique)	ANNUAL	Département	8 vars
ADEME	ADEME (Agence de la Transition Écologique)	ANNUAL	Département	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MTEF	Ministère de la Transition Écologique	ANNUAL	Région	2 vars
CEREMA	Cerema (Centre d'études risques)	ANNUAL	Département	1 var
DGFIP	DGFIP Fiscal Statistics	ANNUAL	Département	1 var
IRSN	IRSN Nuclear Safety	ANNUAL	Département	1 var
ONPES	ONPES / DREES Poverty Observatory	ANNUAL	Département	2 vars
ENEDIS	Enedis Open Data	ANNUAL	Département	2 vars
ANCT	ANCT Observatoire des Territoires	ANNUAL	Département	2 vars
CRE-M	CRE Monitoring Report	ANNUAL	DSO-level	2 vars
CGET	CGET/ANCT Spatial Planning	ANNUAL	Département	1 var
OFCE	OFCE Economic Research	ANNUAL	Région	1 var

ADEME - R	ADEME Régions	ANNUAL	Région	1 var
BRGM	BRGM Geological Survey	STATIC	Département	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE - 1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	COG INSEE Département Registry	STATIC	Département	1 var
ISO - 9223	ISO 9223 Corrosion	DERIVED	Département	1 var
RTE	RTE Open Data (éCO2mix)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF - 0M	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D - 0M	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **7,378 substation scores remain unchanged** this edition. The fleet median R of 0.644 (mean 0.633) reflects a distribution skewed toward the lower bands, with 3.7% in Low and 20.3% in Medium. The first material score changes are expected when the national regulator publishes

the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	§5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	§5
V42-7	NEW	Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	§7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 7,378 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

532

150 EHV · 382 HV

HIGH BAND

5 + 527

100.0% of corridor

CORRIDOR MEDIAN R

0.896

Fleet median: 0.644

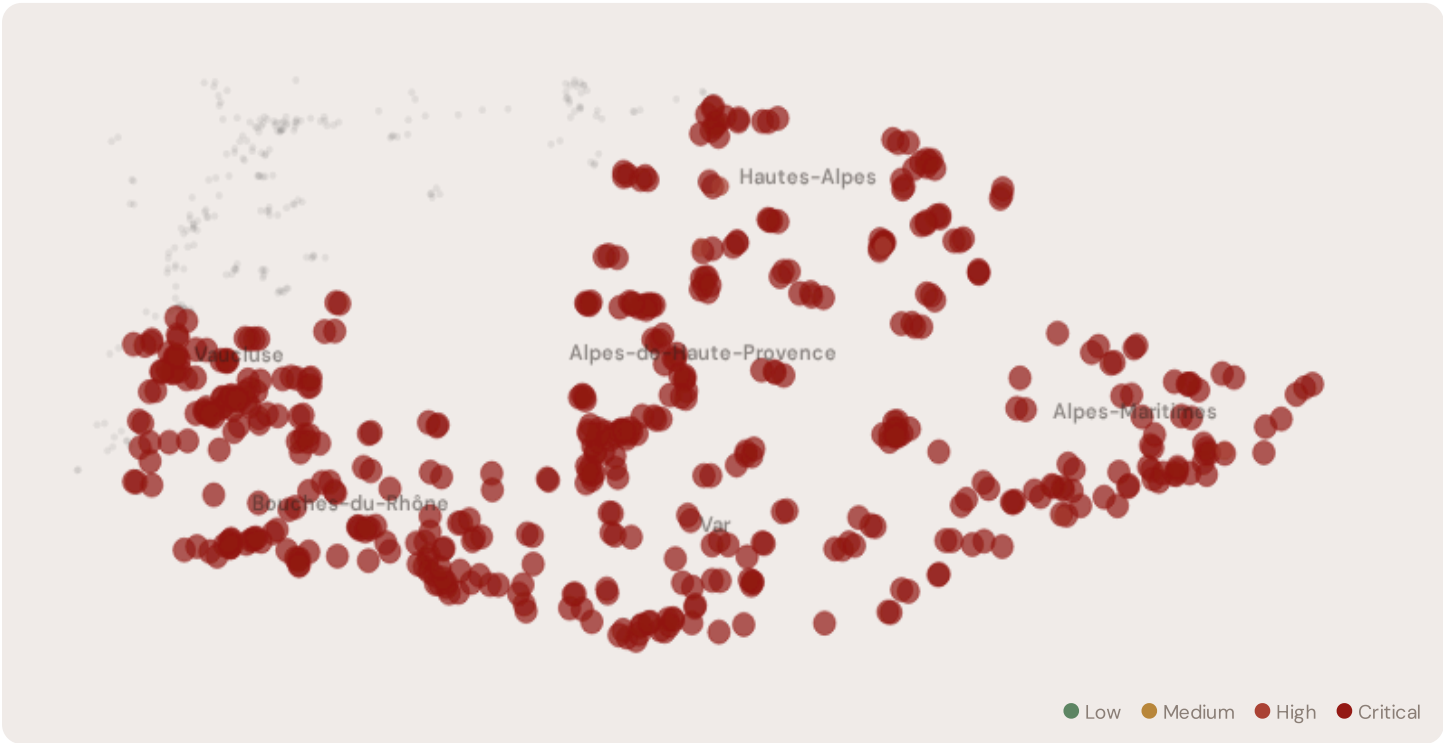
WORST PROVINCE

Var

Avg R: 0.943 · 90 substations

The Provence–Alpes–Côte d’Azur hosts **532 substations** across 6 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **5 substations (0.9%)** are classified High and **527 Critical**, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.644. The corridor median R of 0.896 reflects the local risk profile. The corridor concentrates the country’s industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
Poste Alpes-de- Haute- Provence 073	Alpes- de- Haute- Provence	0.975	Critical	1.000	0.878	0.631	0.000	1.000	0.652	C, V, S	
Poste Alpes-de- Haute- Provence 003	Alpes- de- Haute- Provence	0.974	Critical	0.909	0.888	0.780	0.073	1.000	0.713	V, S	
Poste Alpes-de- Haute- Provence 076	Alpes- de- Haute- Provence	0.967	Critical	0.975	0.912	0.667	0.074	1.000	0.620	V, S	
Poste Var 046	Var	0.965	Critical	1.000	0.604	1.000	0.687	0.292	0.784	C,I	
Poste Var 042	Var	0.964	Critical	0.923	0.632	1.000	0.839	0.244	0.810	I	
Poste Bouches- du-Rhône 008	Bouches- du-Rhône	0.963	Critical	0.841	0.519	1.000	1.000	0.292	0.741	I	
Poste Alpes-de- Haute- Provence 057	Alpes- de- Haute- Provence	0.963	Critical	0.988	0.929	0.489	0.061	1.000	0.644	V, S	
Poste Var 081	Var	0.963	Critical	0.895	0.595	1.000	0.761	0.259	0.906	I,T	
Poste Vaucluse 031	Vaucluse	0.963	Critical	0.871	0.671	0.962	0.321	0.471	0.746		
Poste Var 077	Var	0.962	Critical	0.898	0.673	1.000	0.694	0.297	0.788	I	
... 522 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — PROVENCE-ALPES-CÔTE D'AZUR VS FLEET

C — Continuity	0.838 vs 0.630 (+33%)
I — Infrastructure	0.723 vs 0.346 (+109%)
S — Saturation	0.457 vs 0.356 (+28%)
V — Voltage	0.704 vs 0.670 (+5%)
E — Economic	0.496 vs 0.432 (+15%)
T — Transition	0.705 vs 0.625 (+13%)

MODIFIER AVERAGES — PROVENCE-ALPES-CÔTE D'AZUR VS FLEET

R3 Consequence	1.104	fleet 1.091	↑
R4 Graph Crit.	1.011	fleet 1.006	↑
R6a Restoration	1.011	fleet 0.991	↑
R6b Seismic	1.033	fleet 1.015	↑
R7 Digital Read.	1.000	fleet 1.015	→

The dominant risk driver across the Provence-Alpes-Côte d'Azur corridor is **T (Transition)**, averaging 0.705 against a fleet average of 0.625 — occupying 1410% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.704 vs fleet 0.670 (704% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.104 (fleet: 1.091), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.033, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Var	90 subs · avg R = 0.943 · H:0 C:90 M:0
Bouches-du-Rhône	102 subs · avg R = 0.923 · H:0 C:102 M:0
Alpes-de-Haute-Provence	87 subs · avg R = 0.900 · H:0 C:87 M:0
Vaucluse	95 subs · avg R = 0.884 · H:0 C:95 M:0
Alpes-Maritimes	73 subs · avg R = 0.832 · H:2 C:71 M:0
Hautes-Alpes	85 subs · avg R = 0.807 · H:3 C:82 M:0

The region regions-level breakdown reveals significant intra-regional variation. **Var** leads with a mean R of 0.943 across 90 substations, while **Hautes-Alpes** scores 0.807 — a spread of 0.135 within a single corridor. The SSI's substation-level granularity reveals that the Provence-Alpes-Côte d'Azur is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

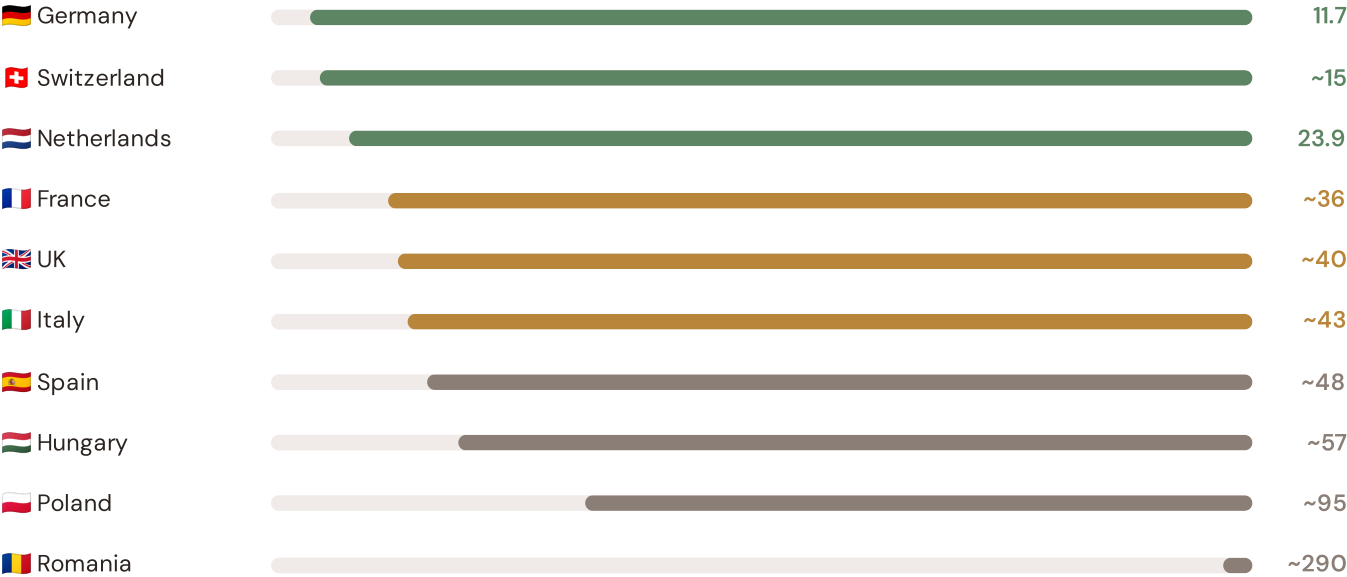
532 Provence-Alpes-Côte d'Azur substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Var sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on France's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



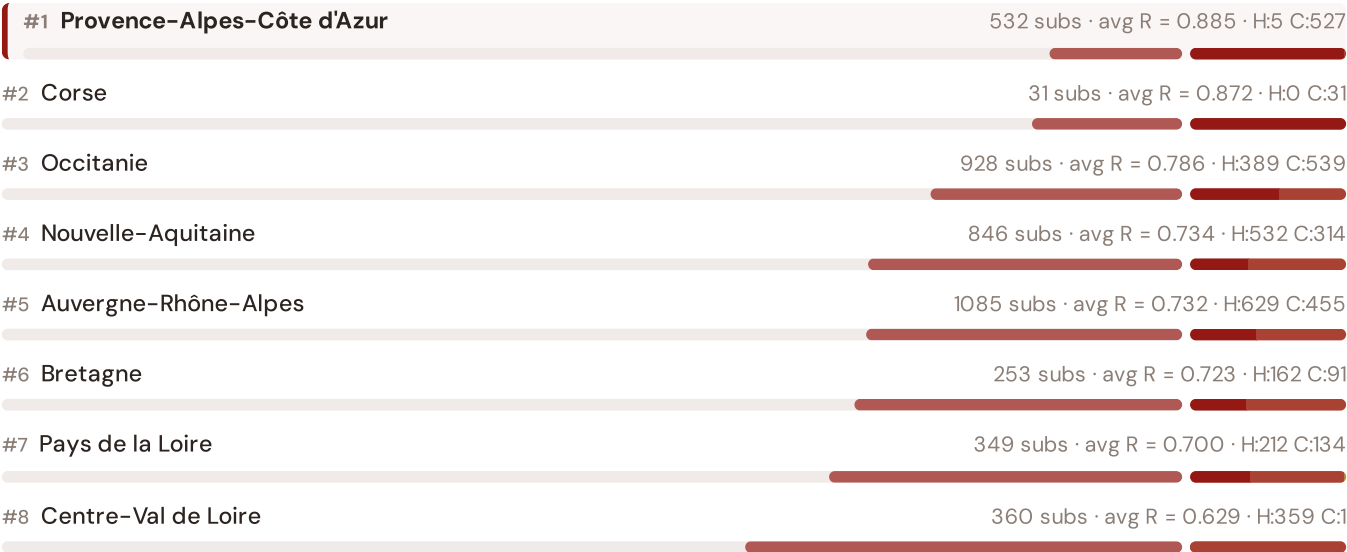
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

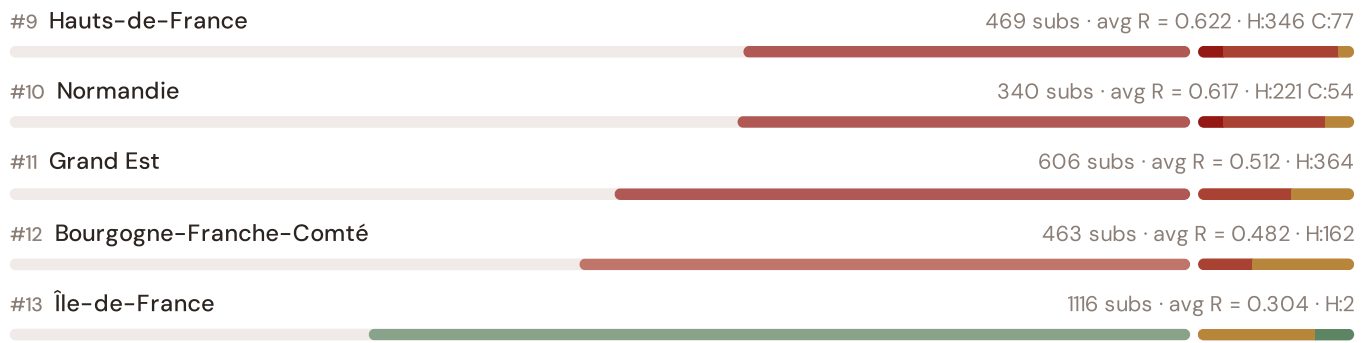
This month's key insight: The Provence-Alpes-Côte d'Azur corridor deep-dive shows **532 substations** (of 532) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.838 is **1.3× the fleet average** of 0.630. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Régions Risk Ranking — All 13 French Régions

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





The Provence-Alpes-Côte d'Azur ranks **#1 of 13 region regions** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed are Provence-Alpes-Côte d'Azur, Corse, Occitanie, while the three most resilient are Île-de-France, Bourgogne-Franche-Comté, Grand Est. 7 of 13 score above the fleet average of 0.633. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

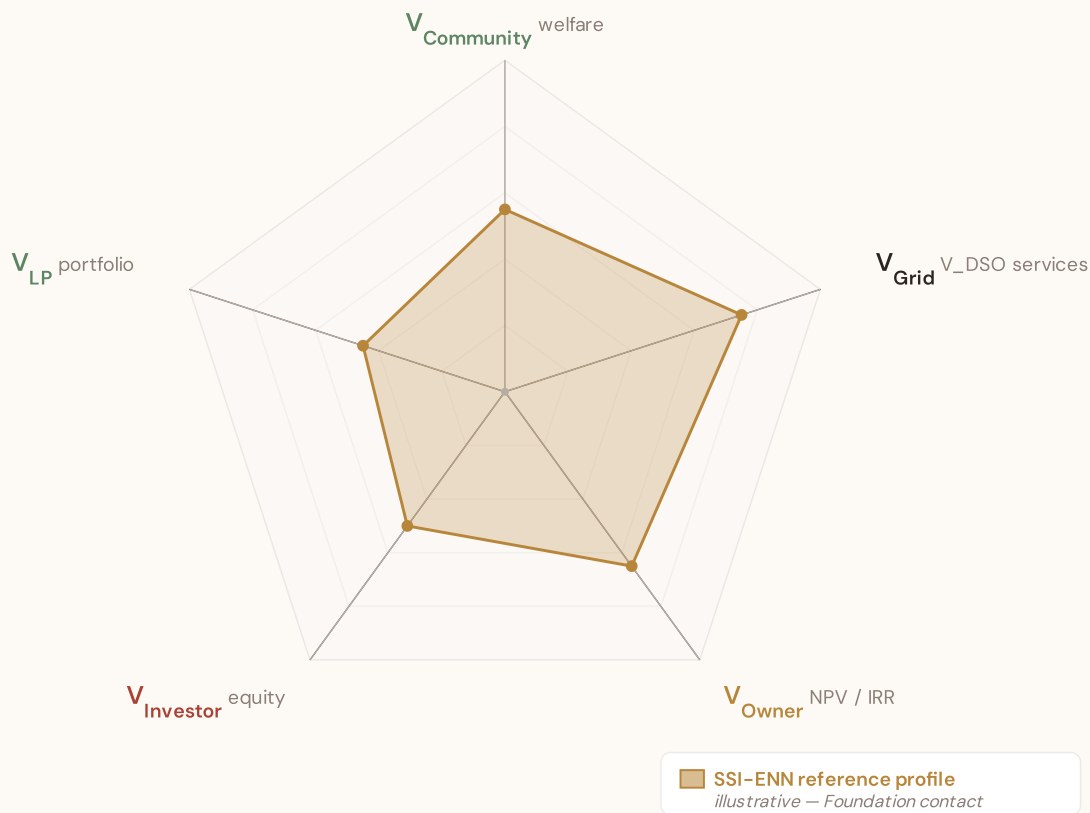
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): $V_{Community}$ (welfare avoided outages, energy-poverty relief, employment) · V_{Grid} (V_{DSO} — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · V_{Owner} (NPV / IRR / payback per asset class) · $V_{Investor}$ (risk-adjusted equity return at the financing tier) · V_{LP} (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal

prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

→ Quantile regression (pinball loss) + conformal prediction

→ Calibrated against SSI Monte Carlo ground truth

→ Per-country, per-voltage, per-band calibration verification

→ Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 7,378 substations with complete feature vectors. Average CI_width of 0.201 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 272 Low, 1500 Medium, 3383 High, 2223 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 2

 Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill	Published-baseline (1) ▲ — solid fill +	Baseline-only (1) — striped fill
W1 · W2 · W3 · W4 · W8	chevron corner marker	W5 — Digital Inclusion
<i>Peer-reviewed methodology operational at fleet scale.</i>	W9 — Supply Security	<i>Foundation methodology under development; baseline value defensible at v4.2.</i>
	<i>Peer-reviewed academic basis</i>	
	<i>(Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).</i>	

Commercial (3) — dashed-faded

W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

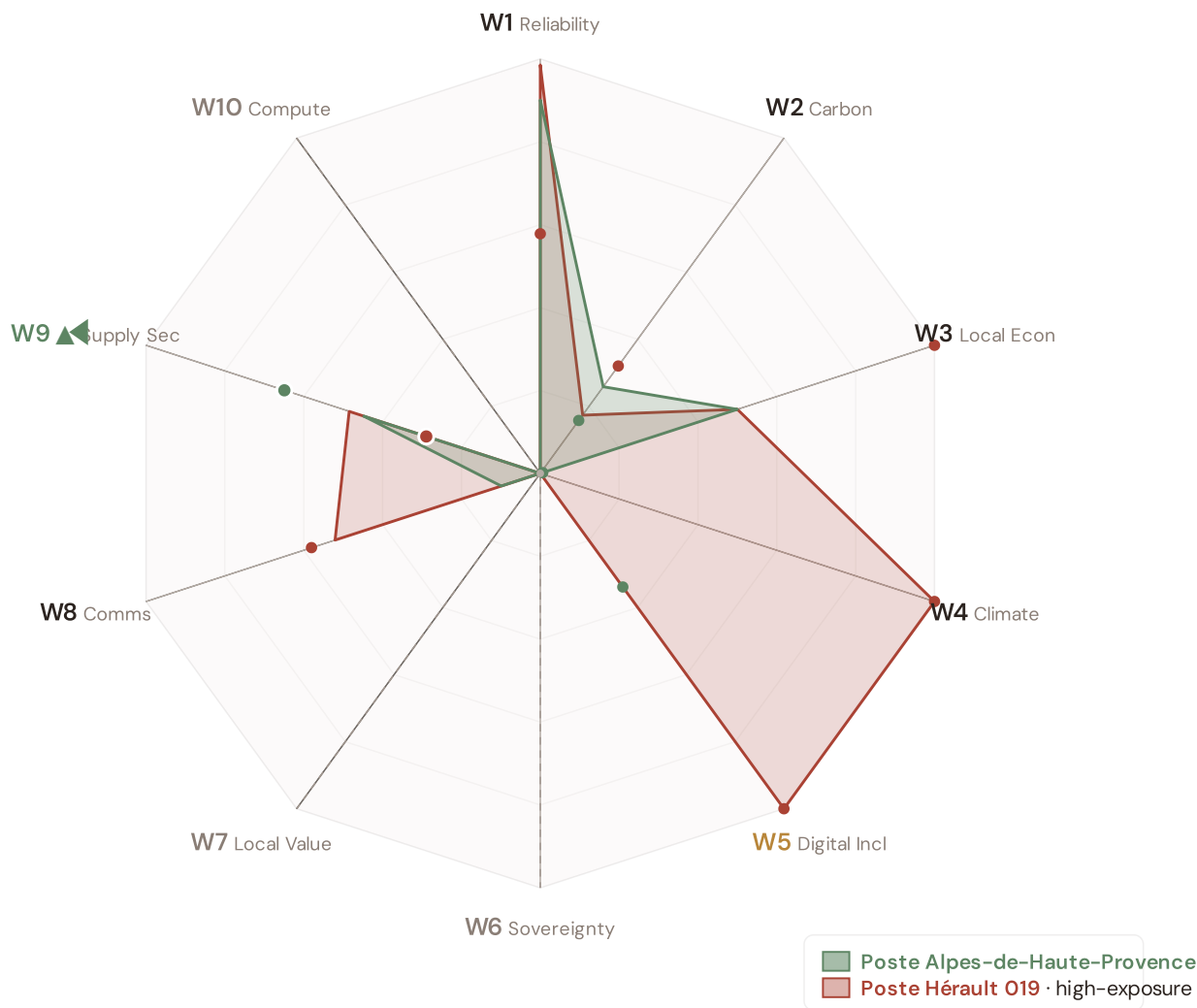
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across France, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Poste Drôme 087** (low-exposure reference, Re_norm 0.000) and **Poste Gard 023** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Poste Alpes-de-Haute-Provence 005 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Poste Hérault 019 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA
COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (\text{SAIDI} + + -\text{POV})$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ \text{EEA})$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$ resilience modifier family
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM} + + \text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population} (\text{Markov 10-yr critical probability})$
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) -$ OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature.

Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — Poste Alpes-de-Haute-Provence 005 (low-exposure reference) and Poste Hérault 019 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected

(min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Capital Metropolitan

Deep-dive into Region 02's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Capital-intensive tier + flood overlay.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CRE (Commission de Régulation de l'Énergie)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

900 Bridge Substations — Single Points of Failure

900 substations are topological bridges — their failure disconnects part of the network. Of these, **693** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Region 02**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an French utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 7,378 substations · 68,379 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.